

C Programming - Days 39 to 49

GitHub-style collection | 2 programs per day | Q77.c to Q98.c

Day 39

Folder structure: **Day-39/** Q77.c | Q78.c

Q77.c

Check if the elements on the diagonal of a matrix are distinct.

C Program

```
#include <stdio.h>

int main() {
    int a[10][10], r, c, i, j, distinct = 1;

    scanf("%d %d", &r, &c);

    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    for (i = 0; i < r && i < c; i++) {
        for (j = i + 1; j < r && j < c; j++) {
            if (a[i][i] == a[j][j]) {
                distinct = 0;
                break;
            }
        }
    }

    if (distinct)
        printf("True");
    else
        printf("False");

    return 0;
}
```

Sample Input

```
3 3
1 2 3
4 5 6
7 8 1
```

Sample Output

```
False
```

Q78.c

Find the sum of main diagonal elements for a square matrix.

C Program

```
#include <stdio.h>

int main() {
    int a[10][10], n, i, j, sum = 0;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        for (j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    for (i = 0; i < n; i++)
        sum += a[i][i];

    printf("%d", sum);

    return 0;
}
```

```
}
```

Sample Input

```
3
1 2 3
4 5 6
7 8 9
```

Sample Output

```
15
```

Day 40

Folder structure: **Day-40/** Q79.c | Q80.c

Q79.c

Perform diagonal traversal of a matrix.

C Program

```
#include <stdio.h>

int main() {
    int a[10][10], r, c, i, j, d;

    scanf("%d %d", &r, &c);

    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);

    for (d = 0; d < r + c - 1; d++) {
        for (i = 0; i < r; i++) {
            j = d - i;
            if (j >= 0 && j < c)
                printf("%d ", a[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Sample Input

```
3 3
1 2 3
4 5 6
7 8 9
```

Sample Output

```
1 2 4 3 5 7 6 8 9
```

Q80.c

Multiply two matrices.

C Program

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], c[10][10] = {0};
    int r1, c1, r2, c2, i, j, k;

    scanf("%d %d", &r1, &c1);
    for (i = 0; i < r1; i++)
        for (j = 0; j < c1; j++)
            scanf("%d", &a[i][j]);

    scanf("%d %d", &r2, &c2);
    for (i = 0; i < r2; i++)
        for (j = 0; j < c2; j++)
            scanf("%d", &b[i][j]);

    if (c1 != r2) {
        printf("Multiplication not possible");
        return 0;
    }

    for (i = 0; i < r1; i++)
        for (j = 0; j < c2; j++)
```

```
        for (k = 0; k < c1; k++)
            c[i][j] += a[i][k] * b[k][j];

    for (i = 0; i < r1; i++) {
        for (j = 0; j < c2; j++)
            printf("%d ", c[i][j]);
        printf("\n");
    }

    return 0;
}
```

Sample Input

```
2 2
1 2
3 4
2 2
5 6
7 8
```

Sample Output

```
19 22
43 50
```

Day 41

Folder structure: **Day-41/** Q81.c | Q82.c

Q81.c

Count characters in a string without using built-in length functions.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int count = 0;

    fgets(str, sizeof(str), stdin);

    while (str[count] != '\0' && str[count] != '\n')
        count++;

    printf("%d", count);

    return 0;
}
```

Sample Input

Hello

Sample Output

5

Q82.c

Print each character of a string on a new line.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0' && str[i] != '\n') {
        printf("%c\n", str[i]);
        i++;
    }

    return 0;
}
```

Sample Input

Hello

Sample Output

H
e
l
l
o

Day 42

Folder structure: **Day-42/** Q83.c | Q84.c

Q83.c

Count vowels and consonants in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100], ch;
    int i = 0, vowels = 0, consonants = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        ch = str[i];

        if (ch >= 'A' && ch <= 'Z')
            ch = ch + 32;

        if (ch >= 'a' && ch <= 'z') {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
                vowels++;
            else
                consonants++;
        }
        i++;
    }

    printf("Vowels = %d\nConsonants = %d", vowels, consonants);
    return 0;
}
```

Sample Input

Hello World

Sample Output

Vowels = 3
Consonants = 7

Q84.c

Convert a lowercase string to uppercase without using built-in functions.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
        i++;
    }

    printf("%s", str);
    return 0;
}
```

Sample Input

```
hello world
```

Sample Output

```
HELLO WORLD
```

Day 43

Folder structure: **Day-43/** Q85.c | Q86.c

Q85.c

Reverse a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0, len = 0;

    fgets(str, sizeof(str), stdin);

    while (str[len] != '\0' && str[len] != '\n')
        len++;

    for (i = len - 1; i >= 0; i--)
        printf("%c", str[i]);

    return 0;
}
```

Sample Input

Hello

Sample Output

olleH

Q86.c

Check if a string is a palindrome.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i, len = 0, palindrome = 1;

    fgets(str, sizeof(str), stdin);

    while (str[len] != '\0' && str[len] != '\n')
        len++;

    for (i = 0; i < len / 2; i++) {
        if (str[i] != str[len - i - 1]) {
            palindrome = 0;
            break;
        }
    }

    if (palindrome)
        printf("Palindrome");
    else
        printf("Not palindrome");

    return 0;
}
```

Sample Input

madam

Sample Output

Palindrome

Day 44

Folder structure: **Day-44/** Q87.c | Q88.c

Q87.c

Count spaces, digits, and special characters in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100], ch;
    int i = 0, spaces = 0, digits = 0, special = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0' && str[i] != '\n') {
        ch = str[i];

        if (ch == ' ')
            spaces++;
        else if (ch >= '0' && ch <= '9')
            digits++;
        else if (!((ch >= 'a' && ch <= 'z') ||
            (ch >= 'A' && ch <= 'Z')))
            special++;

        i++;
    }

    printf("Spaces = %d\nDigits = %d\nSpecial = %d",
        spaces, digits, special);

    return 0;
}
```

Sample Input

Hello 123!

Sample Output

Spaces = 1
Digits = 3
Special = 1

Q88.c

Replace spaces with hyphens in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        if (str[i] == ' ')
            str[i] = '-';
        i++;
    }

    printf("%s", str);
    return 0;
}
```

Sample Input

Hello World C

Sample Output

Hello-World-C

Day 45

Folder structure: **Day-45/** Q89.c | Q90.c

Q89.c

Count frequency of a given character in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100], ch;
    int i = 0, count = 0;

    fgets(str, sizeof(str), stdin);
    scanf("%c", &ch);

    while (str[i] != '\0') {
        if (str[i] == ch)
            count++;
        i++;
    }

    printf("%d", count);
    return 0;
}
```

Sample Input

```
banana
a
```

Sample Output

```
3
```

Q90.c

Toggle case of each character in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        if (str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
        else if (str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
        i++;
    }

    printf("%s", str);
    return 0;
}
```

Sample Input

```
HeLLo
```

Sample Output

```
hEllO
```

Day 46

Folder structure: **Day-46/** Q91.c | Q92.c

Q91.c

Remove all vowels from a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100], ch;
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0') {
        ch = str[i];

        if (!(ch == 'a' || ch == 'e' || ch == 'i' ||
              ch == 'o' || ch == 'u' || ch == 'A' ||
              ch == 'E' || ch == 'I' || ch == 'O' ||
              ch == 'U'))
            printf("%c", ch);

        i++;
    }

    return 0;
}
```

Sample Input

Hello World

Sample Output

Hll Wrld

Q92.c

Find the first repeating lowercase alphabet in a string.

C Program

```
#include <stdio.h>

int main() {
    char str[100];
    int freq[26] = {0};
    int i = 0;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0' && str[i] != '\n') {
        if (str[i] >= 'a' && str[i] <= 'z') {
            freq[str[i] - 'a']++;

            if (freq[str[i] - 'a'] == 2) {
                printf("%c", str[i]);
                return 0;
            }
        }
        i++;
    }

    printf("No repeating character");
    return 0;
}
```

Sample Input

programming

Sample Output

r

Day 47

Folder structure: **Day-47/** Q93.c | Q94.c

Q93.c

Check if two strings are anagrams of each other.

C Program

```
#include <stdio.h>

int main() {
    char a[100], b[100];
    int count[256] = {0};
    int i = 0;

    scanf("%s", a);
    scanf("%s", b);

    while (a[i] != '\0') {
        count[(unsigned char)a[i]]++;
        i++;
    }

    i = 0;
    while (b[i] != '\0') {
        count[(unsigned char)b[i]]--;
        i++;
    }

    for (i = 0; i < 256; i++) {
        if (count[i] != 0) {
            printf("Not anagrams");
            return 0;
        }
    }

    printf("Anagrams");
    return 0;
}
```

Sample Input

```
listen
silent
```

Sample Output

```
Anagrams
```

Q94.c

Find the longest word in a sentence.

C Program

```
#include <stdio.h>

int main() {
    char str[200], longest[100], word[100];
    int i = 0, j = 0, max = 0, k;

    fgets(str, sizeof(str), stdin);

    while (1) {
        if (str[i] != ' ' && str[i] != '\0' && str[i] != '\n') {
            word[j++] = str[i];
        } else {
            if (j > max) {
                max = j;
                for (k = 0; k < j; k++)

```

```

        longest[k] = word[k];
        longest[j] = '\0';
    }
    j = 0;
}

if (str[i] == '\0' || str[i] == '\n')
    break;
i++;
}

printf("%s", longest);
return 0;
}

```

Sample Input

C programming is interesting

Sample Output

programming

Day 48

Folder structure: **Day-48/** Q95.c | Q96.c

Q95.c

Check if one string is a rotation of another.

C Program

```
#include <stdio.h>
#include <string.h>

int main() {
    char a[100], b[100], temp[200];

    scanf("%s", a);
    scanf("%s", b);

    if (strlen(a) != strlen(b)) {
        printf("Not rotation");
        return 0;
    }

    strcpy(temp, a);
    strcat(temp, a);

    if (strstr(temp, b) != NULL)
        printf("Rotation");
    else
        printf("Not rotation");

    return 0;
}
```

Sample Input

```
abcde
deabc
```

Sample Output

```
Rotation
```

Q96.c

Reverse each word in a sentence without changing the word order.

C Program

```
#include <stdio.h>

int main() {
    char str[200];
    int i = 0, start, end, j;

    fgets(str, sizeof(str), stdin);

    while (str[i] != '\0' && str[i] != '\n') {
        while (str[i] == ' ') {
            printf(" ");
            i++;
        }

        start = i;
        while (str[i] != ' ' && str[i] != '\0' && str[i] != '\n')
            i++;

        end = i - 1;
        for (j = end; j >= start; j--)
            printf("%c", str[j]);

        printf(" ");
    }
}
```

```
    return 0;  
}
```

Sample Input

Hello World

Sample Output

olleH dlrow

Day 49

Folder structure: **Day-49/** Q97.c | Q98.c

Q97.c

Print the initials of a name.

C Program

```
#include <stdio.h>

int main() {
    char name[100];
    int i = 0;

    fgets(name, sizeof(name), stdin);

    if (name[0] != ' ')
        printf("%c", name[0]);

    while (name[i] != '\0') {
        if (name[i] == ' ' && name[i + 1] != ' ' &&
            name[i + 1] != '\0' && name[i + 1] != '\n')
            printf("%c", name[i + 1]);
        i++;
    }

    return 0;
}
```

Sample Input

Ansh Kumar Rana

Sample Output

AKR

Q98.c

Print initials of a name with the surname displayed in full.

C Program

```
#include <stdio.h>

int main() {
    char name[100];
    int i = 0, lastSpace = -1;

    fgets(name, sizeof(name), stdin);

    while (name[i] != '\0' && name[i] != '\n') {
        if (name[i] == ' ')
            lastSpace = i;
        i++;
    }

    if (lastSpace == -1) {
        printf("%s", name);
        return 0;
    }

    printf("%c. ", name[0]);

    for (i = 1; i < lastSpace; i++) {
        if (name[i] == ' ' && name[i + 1] != ' ')
            printf("%c. ", name[i + 1]);
    }

    i = lastSpace + 1;
```

```
while (name[i] != '\0' && name[i] != '\n') {  
    printf("%c", name[i]);  
    i++;  
}  
  
return 0;  
}
```

Sample Input

Ansh Kumar Rana

Sample Output

A. K. Rana